



Effect of tip clearance on the heat transfer and pressure drop performance in the micro-reactor with micro-pin-fin arrays at low Reynolds number



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ABSTRACT

To optimize and improve the thermal and hydrodynamic performance of micro-reactors with micro-pin-fin arrays (MPFAR), the effect of tip clearance on the performance of the heat transfer and pressure drop at low Reynolds number was investigated by computational fluid dynamics (CFD). Unlike with high Reynolds number flow, it was found that the heat transfer and pressure drop performance in the case of low Reynolds number is quite sensitive to the tip clearance, and the introduced tip clearance can enhance the heat transfer and reduce the pressure drop effectively. Previous correlations were examined and found to overpredict the average Nusselt number and friction factor. New heat transfer and friction factor correlations accounting for the effect of tip clearance were developed, providing a valuable approach to the design and optimization of the tip clearance of the MPFAR. Finally, a series of experiments was carried out to validate the numerical simulation approach, and the experimental results agreed well with those from simulation.

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1. Introduction

With the rapid application of micro-reactors and electronic microprocessors, there is an increasing need for greater availability of highly efficient technologies that can enhance heat and mass transfer. Among many possible micro-scale structures, micro-pin-fin arrays are commonly used to enhance heat and mass transfer in a variety of applications by increasing the heat transfer surface area and channel flow turbulence [1–3]. In our previous studies, a novel micro-reactor with micro-pin-fin arrays (MPFAR) was designed for hydrogen production, via methanol steam reforming due to the intensification of the heat and mass transfer [4,5]. To enhance energy density, a complete understanding of the performance of the flow characteristics, especially heat transfer, is essential for the design and optimization of micro-reactors [6,7].

Numerous studies have been carried out on the effects of pin-fins on the heat transfer and pressure drop performance in a channel, and a lot of experimental data and empirical correlations have been presented. Short et al. [8,9] studied heat transfer and pressure drop in macro/conventional-sized pin-fin arrays. They proposed new heat transfer and friction factor correlations for two flow regimes of low Reynolds number ($175 < Re < 1000$) and

high Reynolds number ($Re > 1000$). Kosar et al. [10] experimentally investigated the pressure drop characteristics across four arrays of aligned and staggered circular and diamond-shaped micro-pin-fins over Reynolds numbers ranging from 5 to 128, and proposed a new friction factor correlation. Peles et al. [11] theoretically and experimentally analyzed heat transfer and pressure drop over a bank of cylindrical micro-pin-fin arrays. Their work focused mainly on the effect of geometrical and thermo-hydraulic parameters on the total thermal resistance. Kosar and Peles [12,13] studied heat transfer and pressure drop in an array of staggered circular pin-fins. Water and the refrigerant R-123 were used as working liquids. Previous correlations overpredicted the Nusselt number at low Reynolds number in their study, so they proposed a new heat transfer correlation based on the water and R-123 data. Qu and Siu-Ho [14,15] studied the heat transfer and pressure drop characteristics of liquid single-phase flow in an array of micro-pins. In their studies, two inlet temperatures and six maximum mass velocities were tested in an array of 1950 staggered square micro-pin-fins. New correlations were then developed to predict the heat transfer and pressure drop performance in the heat sink of the micro-pin-fin arrays. Tullius et al. [16] studied the effects of pin-fin shapes, channel clearances, fin material and the geometry of short micro-pin-fin arrays through numerical simulation using CFD. Correlations describing the Nusselt number and friction factor were obtained, which applied only to short fins in a laminar regime.

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Nomenclature

A_b	surface area of the heated base region	Nu_{ave}	average Nusselt number
A_c	cross section area of a micro-pin-fin	$Nu_{ave,pred}$	predicted average Nusselt number
A_{fin}	wetted surface area of a micro-pin-fin	$Nu_{ave,sim}$	simulated average Nusselt number
A_{min}	minimum transverse flow area of the micro channel	P_{fin}	cross section perimeter of a micro-pin-fin
A_t	total area of the micro-pin-fin arrays base	P_{in}	inlet pressure
C	tip clearance	P_{out}	outlet pressure
c, c_f	coefficient in heat transfer and friction factor correlations, respectively	Pr	Prandtl number
c_p	specific heat	$Pr_{f,ave}$	Prandtl number at $T_{f,ave}$
D_c	hydraulic diameter of the channel	Q	heat transferred to the micro-pin-fin arrays
D_f	hydraulic diameter of a micro-pin-fin	Q_{loss}	heat loss
f	flow friction factor	q	heat flux applied on the base of the micro-pin-fin arrays
$f_{ave,pred}$	predicted average flow friction factor	Q_f	volume flow rate
$f_{ave,sim}$	simulated average flow friction factor	Re	Reynolds number based on the pin-fin size
h	heat transfer coefficient	Re_c	Reynolds number based on the channel size
h_{ave}	average heat transfer coefficient of the micro-pin-fin arrays	S_L	longitudinal pitch (center to center)
h_b	height of the base	S_T	transverse pitch (center to center)
h_{base}	heat transfer coefficient of the base	T	temperature
h_c	height of the channel	$T_{f,ave}$	average temperature of water
h_f	height of a pin-fin	T_{in}	inlet temperature of water
h_{fin}	heat transfer coefficient of the micro-pin-fin	T_{nw}	volume averaging temperature of near wall water
I, J	the exponents in friction factor correlation	T_{out}	outlet temperature of water
j, m, n	the exponents in friction factor correlation	T_w	wall temperature
k	thermal conductivity	$T_{w,ave}$	average temperature of channel bottom
L	length of the micro-pin-fin arrays	u	flow velocity
L_c	length of the micro channel	u_{in}	inlet flow velocity
l_d	distance between the four thermocouples and top surface	u_{max}	maximum flow velocity in the micro-pin-fin arrays
M	number of data points	W	width of micro-pin-fin array
m_{fin}	fin parameter	ΔP	pressure drop
$\dot{m}$	mass flow rate	Greek symbols	
N	total number of micro-pin-fins	Λ	interface between the solid and fluid regions
N_L	row number of micro-pin-fins	η_{fin}	fin efficiency
N_T	column number of micro-pin-fins	μ	viscosity
Nu	Nusselt number	ρ	density

However, most of the studies mentioned above were focused mainly on the operating conditions and the geometry of micro-pin-fin arrays. Little attention has been paid to the effect of tip clearance. Reyes et al. [17] studied the effect of tip clearance in micro channels on the performance of heat transfer and pressure drop, and optimized the tip clearance accounting for both thermal and pressure drop performance based on the experimental results. Min et al. [18] found that when the tip clearance was about 40% of the channel height, the heat transfer and pressure drop performance was better. Moores et al. [19,20] studied the effect of tip clearance on the thermal and hydrodynamic performance of staggered pin-fin arrays. In their study, the tip clearance ranged from 0 to 25% of pin-fin height with a Reynolds number range of 200 to 1000. The introduction of a small tip clearance was found to enhance the heat transfer performance in some cases due to the additional surface area that was exposed to the fluid flow. More recently, Rozati et al. [21] studied heat transfer in shrouded pin-fin arrays with and without tip clearance through numerical analysis. Their study was focused mainly on the effects of tip clearance on the heat transfer performance. They found some reasonable values of tip clearance that provided the best performance under certain conditions.

Most of the previous studies about the tip clearance in micro-channels were carried out at high Reynolds number, while the Reynolds number of the reagent flowing into micro-reactors for chemical reactions is very small (normally $Re < 300$). Heat transfer is

quite sensitive to the tip clearance at low Reynolds number. Previous correlations may not be suitable for describing the heat transfer and pressure drop performance in the MPFAR at low Reynolds number.

Therefore, the aim of this study was to investigate the effect of different tip clearances on the heat transfer and pressure drop performance of the MPFAR at low Reynolds numbers. A computational model of the MPFAR with a tip clearance range of 0 μm to 1000 μm was developed. The thermal and hydrodynamic performance was studied by numerical analysis at low Reynolds numbers, ranging from 20 to 350. A series of experiments was then carried out to validate the computational model in the numerical analysis. The average Nusselt number and friction factor data obtained were compared to the predictions of previous correlations. Finally, new heat transfer and friction factor correlations accounting for the effect of tip clearance were developed to provide a valuable approach for the design and optimization of the MPFAR.

2. Computational modeling

The computational model of the MPFAR is shown schematically in Fig. 1. The model is focused on forced flow passing through micro-pin-fin array reactors with or without tip clearance. Tip clearance (C) can be defined as the subtraction of the pin-fin height (h_f) from the channel height (h_c). The detailed dimensions of the models are listed in Table 1. In the present study, different tip

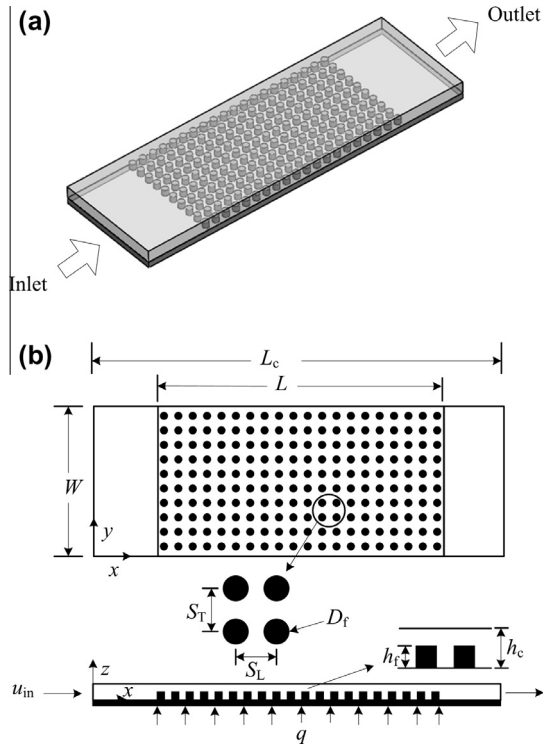


Fig. 1. Computational model: (a) 3-D simulation model of the MPFAR; (b) schematics of the geometrical structure.

clearances were obtained by varying the channel height with the pin-fin height held constant, so that the tip clearance ranged from 0 μm to 1000 μm . There were 10×50 pin-fins in line arrays on rectangular minichannels with dimensions ($W \times L \times h_c$) of 18 mm $\times$ 90 mm $\times$ 1 mm, 18 mm $\times$ 90 mm $\times$ 1.2 mm, 18 mm $\times$ 90 mm $\times$ 1.5 mm and 18 mm $\times$ 90 mm $\times$ 2 mm. The transverse and longitudinal distances between pin-fins (center to center), S_T and S_L , were kept constant for all the simulations and were equal to 1.8 times the diameter of the pin-fin (D_f). The pin-fin height was the same as the diameter of the pin-fin in all five geometrical models.

In analyzing the problem, the flow was assumed to be laminar, steady-state and incompressible [22–25]. The Navier–Stokes and energy equations for the flow are as follows:

Continuity equation for the fluid region

$$\nabla \cdot \rho_f u = 0 \quad (1)$$

Momentum equation for the fluid region

$$\rho_f (u \cdot \nabla u) = -\nabla P + \mu_f \nabla^2 u \quad (2)$$

Energy equation for the fluid region

$$\rho_f c_p (u \cdot \nabla T) = k_f \nabla^2 T \quad (3)$$

Energy equation for the solid region

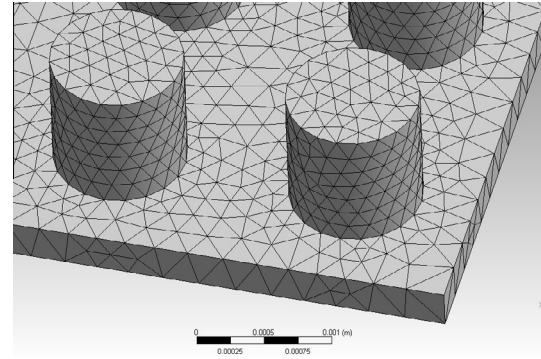


Fig. 2. Grid in the vicinity of the pin-fin.

$$k_s \nabla^2 T = 0 \quad (4)$$

As for the thermal boundary conditions, an idealized uniform heat flux was assumed at the bottom surface of the micro-pin-fin array substrate, expressed as:

$$q = -k_s \frac{\partial T}{\partial z}, \quad \text{at } \frac{L_c - L}{2} \leq x \leq \frac{L_c + L}{2}, \quad 0 \leq y \leq W, \quad z = -h_b \quad (5)$$

All other boundaries except the bottom surface could be considered as adiabatic. The continuities of temperature and heat flux of the model were coupled at the interface between the solid and fluid regions, which can be expressed as:

$$T_{s,\Lambda} = T_{f,\Lambda} \quad (6)$$

$$-k_s \frac{\partial T_s}{\partial n} \Big|_{\Lambda} = -k_f \frac{\partial T_f}{\partial n} \Big|_{\Lambda} \quad (7)$$

where Λ represents the interface between the solid and fluid regions.

As for the hydraulic boundary conditions, a uniform velocity was applied at the channel inlet,

$$u = u_{in}, \quad v = 0, \quad w = 0, \quad \text{at } x = 0, \quad 0 \leq y \leq W, \quad 0 \leq z \leq h_c \quad (8)$$

Considering the geometrical and physical symmetry, the structure of the model was simplified as only one half of the real model shown in Fig. 1(a). The micro-pin-fin arrays were made of the aluminum alloy A356 and deionized water was used as the working fluid. The inlet temperature and the outlet static pressure were kept constant at 298.15 K and 0 Pa, respectively, for all simulations. The multiple mass flow rates at the inlet ranged from 1 to 25 L/h covering all the needed inlet velocities that varied from 0.008 to 0.2 m/s. The numerical model was then meshed into more than one million tetrahedral elements automatically and typical size of the elements was $9\text{e-}6$ m, as shown in Fig. 2. The SIMPLE algorithm was used to solve the governing equations. All equations were numerically solved iteratively using the commercial fluid dynamics program ANSYS FLUENT 12.1.2 (2009, ANSYS, Inc.). The solution was regarded as converged when the residuals for both

Table 1
Detailed dimensions of the MPFAR.

No.	D_f (mm)	S_L (mm)	S_T (mm)	N	W (mm)	L (mm)	h_c (mm)	h_f (mm)	h_c/h_f
1	1	1.8	1.8	10×50	18	90	1	1	1
2	1	1.8	1.8	10×50	18	90	1.1	1	1.1
3	1	1.8	1.8	10×50	18	90	1.2	1	1.2
4	1	1.8	1.8	10×50	18	90	1.5	1	1.5
5	1	1.8	1.8	10×50	18	90	2	1	2

continuity and momentum were less than 10^{-5} . All the simulations were studied in the single phase regime and the fluid properties were simplified as constant.

In order to check the sensitivity of the numerical results, meshes with a total of 7,021,567, 1,037,892 and 1,643,872 elements were employed in the micro reactor with the tip clearance of $h_c/h_f = 2$ to evaluate the grid independence at the highest inlet velocity of 0.15 m/s, which guaranteed the independency of the results at lower inlet velocity as well. The influence of the grids on the Nusselt number and friction factor is shown in Table 2. The deviations of Nusselt number and friction factor are less than 1.4% and 0.8%, respectively. In Table 2, the results show that the numerical analysis based on the mesh with a total of 1,037,892 elements used in our numerical analysis can achieve an acceptable accuracy and is fundamentally grid independent in these cases.

In the simulation study, the maximum flow velocity in the micro-pin–fin arrays is defined as

$$u_{\max} = \frac{Q_f}{A_{\min}} \quad (9)$$

where $A_{\min}$ is the minimum transverse flow area of the channel, and it is calculated from

$$A_{\min} = WLh_c - N_f D_f h_f \quad (10)$$

The Reynolds number in the present study is given by

$$Re = \frac{\rho_f u_{\max} D_f}{\mu_f} \quad (11)$$

where μ_f is the water viscosity at temperature T_{in} .

Heat flux q is applied to the bottom surface of the micro-pin–fin substrate. The heat transfer coefficient is obtained from

$$h = \frac{q}{T_w - T_{nw}} \quad (12)$$

To analyze accurately the heat transfer across the micro-pin–fin arrays, an average heat transfer coefficient for the entire micro-pin–fin array has been employed. The average heat transfer coefficient h_{ave} is determined from

$$h_{ave} = \frac{h_{fin} A_{fin} + h_{base} A_b}{A_{fin} + A_b} \quad (13)$$

where h_{fin} and h_{base} represent the heat transfer coefficients of the micro-pin–fin and the base of the only heated region, calculated by Eq. (12), respectively. A_{fin} and A_b are the wetted surface areas of the micro-pin–fins and the heated base region which the flow fluid touches, respectively.

The average Nusselt number Nu_{ave} is calculated from

$$Nu_{ave} = \frac{h_{ave} D_f}{k_f} \quad (14)$$

The pressure drop ΔP across the micro-pin–fin arrays is obtained from

$$\Delta P = P_{in} - P_{out} \quad (15)$$

Once the ΔP is determined, the flow friction factor f can be calculated using [26]

$$f = \frac{\Delta P}{N_L \left(\frac{\rho_f u_{\max}^2}{2} \right)} \quad (16)$$

Table 2
Nusselt number and friction factor with different grids under the same conditions.

Grids	7,021,567	1,037,892	1,643,872
Nu_{ave}	8.9061	8.78333	8.82687
f	0.21311	0.21329	0.21478

3. Results and discussion

In this section, we describe a numerical simulation of the effect of tip clearance on the heat transfer and pressure drop performance. The results are used to obtain empirical correlations that can predict the thermal and hydrodynamic performance of the MPFAR with different tip clearances.

3.1. The effect of tip clearance on the heat transfer performance

According to Eqs. (11) and (14), we can obtain the variation in the average Nusselt number Nu_{ave} with the Reynolds number Re , as shown in Fig. 3. The tip clearance includes values of h_c/h_f of 1, 1.1, 1.2, 1.5 and 2. Correlations based on the present data are also plotted. The tip clearance evidently enhances the heat transfer performance of the micro-reactor for Reynolds numbers ($33 < Re < 350$), especially when $h_c/h_f = 1.1, 1.2$ or 1.5 . When the Reynolds number is fixed as 100, the average Nusselt number is increased by about 100% with a tip clearance of $h_c/h_f = 1.1$, compared to the one without tip clearance. This is due to that at low levels of tip clearance, the inclusion of tip clearance can induce accelerated flow in the tip clearance, reduce the development of recirculating wakes and redistribute the flow along the height of the channel. However, as the tip clearance is further increased, the effect of fluid bypassing the arrays becomes dominant, and the resulting reduction in flow rate within the arrays produces lower heat transfer performance. Therefore, the average Nusselt number decreases with increasing tip clearance at high levels of tip clearance in this study, as shown in Fig. 3, which agrees well with the results from a study of heat transfer in micro-square pin–fin arrays by Tullius et al. [16]. Fig. 3 also shows that the average Nusselt number for a micro-reactor without tip clearance approaches the maximum value of about 11 when the Reynolds number reaches about 275, after which the increase is not significant. However, for a micro-reactor with tip clearance (especially when $h_c/h_f = 1.1, 1.2$ or 1.5), the rate of increase of the average Nusselt number is still large when the Reynolds number reaches 350, which reveals a significant enhancement in the heat transfer performance due to the tip clearance.

3.2. The effect of tip clearance on the pressure drop performance

Based on Eqs. (11) and (16), we can obtain the relationship between the friction factor f and the Reynolds number Re , as shown in Fig. 4. When the tip clearance of h_c/h_f is decreased from 2 to 1, the friction factor nearly doubles at a fixed Reynolds number of

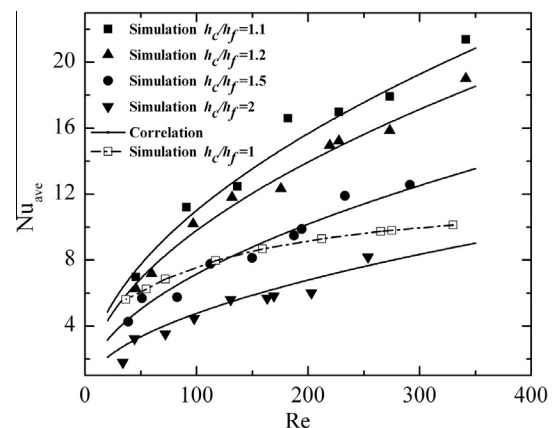


Fig. 3. Variation in the average Nusselt number with Reynolds number under different tip clearances.

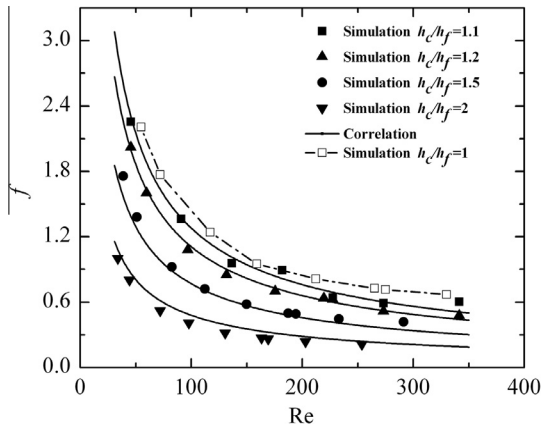


Fig. 4. Variation in the friction factor with Reynolds number under different tip clearances.

100, indicating a large increase in pressure drop across the micro-pin-fin arrays. The introduced tip clearance reduces the flow friction and improves the flow performance in the micro-reactor. Variation in the friction factor with the Reynolds number under different tip clearances shows a similar trend, indicating that the friction factor can be described by a formula with the variable h_c/h_f .

The increase in the tip clearance reduces the heat transfer in the MPFAR, while the pressure drop performance is improved significantly. Therefore, an optimized value of tip clearance can be obtained.

3.3. New heat transfer correlation

Classic heat transfer theory shows that the Nusselt number is a function of the Reynolds number and the Prandtl number in most previous pin-fin heat transfer correlations, and can be written as:

$$Nu = cRe^m Pr_{f,ave}^n \quad (17)$$

where $Pr_{f,ave}$ is the Prandtl number of water at the temperature $T_{f,ave}$. The exponents m and n represent the dependence of Nu on Re and $Pr_{f,ave}$, respectively, and are usually constants in the correlations. The coefficient c is a function of geometrical parameters of the pin-fin arrays to account for the effect of geometrical parameters on heat transfer.

Many heat transfer correlations have been developed by previous studies. Six typical correlations are summarized in Table 3. Correlation 1 developed by Zukauskas [27] and correlation 2 developed by Short et al. [8] are based on experimental heat transfer

data for conventional macro-pin-fin arrays. Correlation 3 from Moores and Joshi [19] and correlation 4 from Tullius et al. [16] are for mini-pin-fin arrays. Correlation 5 proposed by Kosar and Peles [13] and correlation 6 proposed by Qu and Siu-Ho [14] are for micro-pin-fin arrays.

To compare the present average Nusselt number data with the predictions of previous correlations, the mean absolute error (MAE) for each heat transfer correlation is defined as:

$$MAE_{Nu} = \frac{1}{M} \sum \frac{|Nu_{ave,sim} - Nu_{ave,pred}|}{Nu_{ave,sim}} \times 100\% \quad (18)$$

where M is the number of data points.

The average Nusselt number data from this study is compared with the previous correlations predictions as shown in Fig. 5. The MAE values with respect to the present data range from 27.08% to 131.44%, as listed in Table 3. Fig. 5 shows that the previous typical pin-fin array correlations overpredict the present Nu data. Among the six previous correlations, correlation 3 is for micro-pin-fin arrays with tip clearance. However, it overpredicts with an MAE of 131.44. Note that correlation 3 was developed for a high Reynolds number in the range of 200 to 1000, which is not well suited for the present study.

Table 3 and Fig. 5 show that there is obvious discrepancy between the six previous correlation predictions and present data in this study, and the six previous heat transfer correlations presented by other authors cannot describe the heat transfer performance in the MPFAR with sufficient accuracy. There are many possible reasons for the comparatively poor predictive performance of the previous correlations, including: (1) the effect of the geometrical configurations and parameters, including the shape of the pin-fin and especially the tip clearance introduced in the present study. We believe that the introduced tip clearance will significantly affect the heat transfer and pressure drop characteristics, and the most obvious reason is that exposing the tips of the pin-fin to the working fluid increases the total heat transfer area; (2) the effect of the operating conditions, such as the type of working fluid, the fluid flow Reynolds number and the operating temperature. The six previous heat transfer correlations are developed for relative high Reynolds number, while the flow across the micro-pin-fin arrays develops an unsteady Karman wake past each micro-pin-fin and may be turbulent with the increase of Reynolds number. The thermal and hydrodynamic performance of pin-fin arrays at low Reynolds number is very different from that at relatively high Reynolds number. Therefore, a new heat transfer correlation accounting for the effect of tip clearance at low Reynolds number is developed to fit the present data through modifying the common functional forms given by Eq. (17). The term $(h_c/h_f)^{1/3}$,

Table 3
Correlations for heat transfer in pin-fin arrays.

No.	References	Average Nusselt number	Scale	Reynolds number	Clearance range (h_c/h_f)	Pr	h_f/D_f	MAE (%)
1	Zukauskas [27]	$Nu = 0.9Re^{0.4} Pr_{f,ave}^{0.36} \left(\frac{Pr_{f,ave}}{Pr_{w,ave}} \right)^{0.25}$	Macro	10–100	1	0.5–500	>8	46.24
2	Short et al. [8]	$Nu = 0.76 \left(\frac{S_L}{D_f} \right)^{0.16} \left(\frac{S_T}{D_f} \right)^{0.2} \left(\frac{h_c}{D_f} \right)^{-0.11} Re^{0.33} Pr_{f,ave}^{1/3}$	Macro	<1000	1	–	1.88–7.25	36.16
3	Moores and Joshi [19]	$Nu = 0.64 \left(\frac{h_c}{D_f} \right)^{0.36} \left(\frac{h_c}{h_f} \right)^{-0.57} Re^{0.64} Pr_{f,ave}^{0.36}$	Mini	200–1000	1–1.25	5.4–5.7	0.5–1.1	131.44
4	Tullius et al. [16]	$Nu = 0.08 \left(\frac{S_L}{D_f} \right)^{0.2} \left(\frac{S_T}{D_f} \right)^{0.2} \left(\frac{h_c}{D_f} \right)^{0.25} \left(1 + \frac{h_c - h_f}{D_f} \right) Re^{0.6} Pr_{f,ave}^{0.36} \left(\frac{Pr_{f,ave}}{Pr_{w,ave}} \right)^{0.25}$	Mini	50–1500	1.33–4	–	0.25–0.75	51.57
5	Kosar and Peles [13]	$Nu = 0.0423Re^{0.99} Pr_{f,ave}^{0.21} \left(\frac{Pr_{f,ave}}{Pr_{w,ave}} \right)^{0.25}$	Micro	134–314	1	–	2.44	34.41
6	Qu and Siu-Ho [14]	$Nu = 0.0285Re^{0.932} Pr_{f,ave}^{1/3}$	Micro	78.6–179.6	1	–	3.35	47.08
7	Present study	$Nu = 0.75 \left(\frac{h_c}{h_f} \right)^{-1.41} Re^{0.51} Pr_{f,ave}^{1/3}$	Micro	33–350	1–2	5.27–5.76	1	7.55

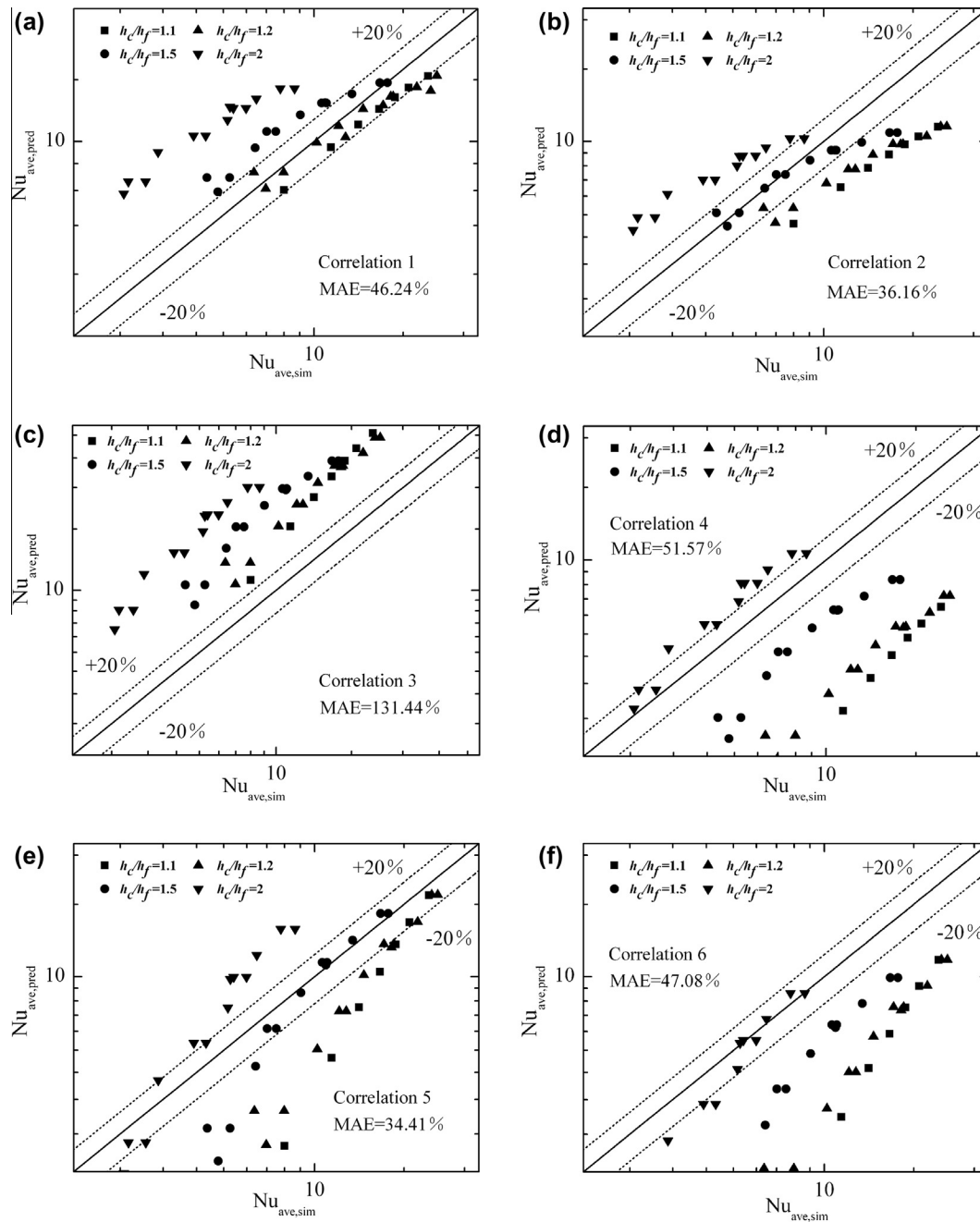


Fig. 5. Comparison of the average Nusselt number data with predictions of six previous heat transfer correlations: (a) correlation 1; (b) correlation 2; (c) correlation 3; (d) correlation 4; (e) correlation 5; (f) correlation 6.

representing the effect of the tip clearance, is added to the function. In the present study, only water was tested as the working fluid, and the range of the Prandtl number of water is not wide enough to correlate the exponent n in Eq. (17). The values of n in the previous correlations (correlations 2 and 6) are both $1/3$ (as shown in Table 3). In the present study, the exponent n is also assumed to be $1/3$. Thus, the new heat transfer correlation can be expressed as:

$$Nu = c \left(\frac{h_c}{h_f} \right)^j Re^m Pr_{f,ave}^{1/3} \quad (19)$$

A non-linear multiple regression analysis is used to determine the values of the coefficient c and the exponents m and j in Eq. (19). Thus, we obtain

$$Nu = 0.75 \left(\frac{h_c}{h_f} \right)^{-1.41} Re^{0.51} Pr_{f,ave}^{1/3} \quad (20)$$

Fig. 6 compares the prediction of the new heat transfer correlation 7 (as listed in Table 3) with respect to the present data. The MAE of 7.55% demonstrates the good predictive capability of the new heat transfer correlation 7 developed in the present study.

3.4. New friction factor correlation

After reviewing many other previous studies for obtaining the friction factor f in the MPFAR, six previous friction factor correlations in pin–fin arrays are selected and listed in Table 4.

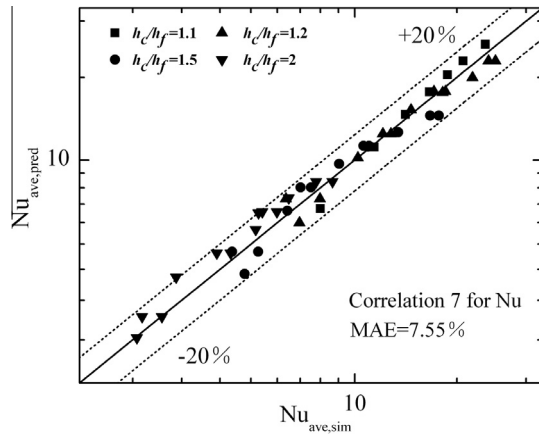


Fig. 6. Comparison of the average Nusselt number data with predictions of the new heat transfer correlation.

The mean absolute error (MAE) for each friction factor correlation is defined as:

$$MAE_f = \frac{1}{M} \sum \frac{|f_{ave,sim} - f_{ave,pred}|}{f_{ave,sim}} \times 100\% \quad (21)$$

where M is the number of data points.

The mean absolute error (MAE) for each friction factor correlation is given in Table 4. From Table 4, it can be seen that the previous friction factor correlations are not able to describe the pressure drop characteristics in the MPFAR with sufficient accuracy. Therefore, a new friction factor correlation is proposed as follows:

$$f = c_f \left(\frac{h_c}{h_f} \right)^J Re^I \quad (22)$$

The term $(h_c/h_f)^J$ represents the effect of the tip clearance on the pressure drop across the micro-pin-fin arrays.

The values of coefficient c_f and the exponents I and J in Eq. (22) are determined through a non-linear multiple regression analysis. Thus, the final friction factor correlation is developed as:

$$f = 42.27 \left(\frac{h_c}{h_f} \right)^{-1.64} Re^{-0.75} \quad (23)$$

Fig. 7 shows a comparison of the present data with the new friction factor correlation 7 (as listed in Table 4). The MAE of the new friction factor correlation 7 is 10.21%, indicating a good agreement with present data in this study.

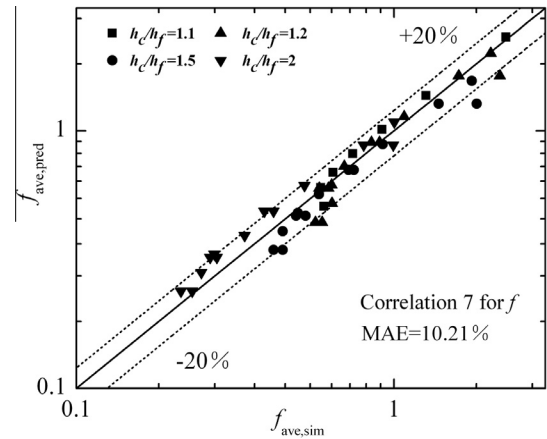


Fig. 7. Comparison of the friction factor data with predictions of the new friction factor correlation.

4. Experimental validation

A series of experiments was conducted to evaluate the heat transfer and pressure drop performance of the MPFAR and to verify the reliability of the numerical analysis. The simulated results obtained in Section 3 were then compared with the experimental results.

4.1. Experimental setup

The experimental setup consisted of a flow loop, a test section and a data acquisition/control system, as shown in Fig. 8. In the flow loop, deionized water was circulated by a gear pump and flowed firstly through a 50 μ m filter to prevent solid particles from clogging the flow channel. The water then flowed through a control valve and flow rotameter (KROHNE, DK800-4), so that a desired flow rate could be set. The rotameter was calibrated with respect to the working fluid by using the standard weighting method. The result showed that the accuracy of the flow rate measurement was better than 4% of its readings. A compact temperature bath was used to keep the inlet at a desired operating temperature. Another temperature bath was used to cool the water before it entered the water reservoir.

The micro-reactor test section is shown in Fig. 9(a), which is composed of a cover plate, a housing and micro-pin-fin arrays. All three parts were made of the aluminum alloy A356. In this study, the micro-pin-fin arrays are shown in Fig. 9(b), which were fabricated by the micro thixo-forming processing. Compared to the conventional micro-fabrication techniques, such as ultra-precision machining and micro-forming, the micro thixo-forming processing

Table 4
Correlations for friction factor in pin-fin arrays.

No.	References	Friction factor	Scale	Reynolds number	Clearance range (h_c/h_f)	h_f/D_f	MAE (%)
1	Gunther and Shaw [28]	$f = \frac{180}{Re} \left(\frac{4S_f S_r}{\pi D_f^2} - 1 \right)^{0.4} \left(\frac{S_r}{S_f} \right)^{0.6}$	Macro	<1000	1	>8	272.47
2	Short et al. [9]	$f = 140.4 \left(\frac{S_r}{D_f} \right)^{-1.3} \left(\frac{S_r}{D_f} \right)^{-0.78} \left(\frac{h_f}{D_f} \right)^{-0.55} Re^{-0.65}$	Macro	<1000	1	1.88–7.25	188.06
3	Moore and Joshi [19]	$f = 19.04 \left(\frac{h_f}{D_f} \right)^{-0.742} \left(\frac{h_c}{h_f} \right)^{0.505} Re^{-0.502}$	Mini	200–1000	1–1.25	0.5–1.1	172.95
4	Prasher et al. [26]	$f = 3.5 \left(\frac{S_r - D_f}{D_f} \right)^{-0.442} \left(\frac{S_r - D_f}{D_f} \right)^{-0.245} \left(\frac{h_f}{D_f} \right)^{0.724} Re^{-0.58}$	Micro	40–1000	1	1.3, 2.48, 2.8	60.06
5	Kosar et al. [10]	$f = \frac{2259}{Re^{1.7}} \left(\frac{h_f/D_f}{h_f/D_f + 1} \right)^{1.9} \left(\frac{S_r S_f}{A_c} \right)^{-0.4} + \frac{54}{Re^{0.7}} \left(\frac{1}{h_f/D_f + 1} \right)^2 \left(\frac{S_r S_f}{A_c} \right)^{-0.7}$	Micro	5–128	1	1, 2	56.29
6	Qu and Siu-Ho [15]	$f = 20.09 Re^{-0.547}$	Micro	37.9–85.8	1	3.35	131.06
7	Present study	$f = 42.27 \left(\frac{h_c}{h_f} \right)^{-1.64} Re^{-0.75}$	Micro	33–350	1–2	1	10.21

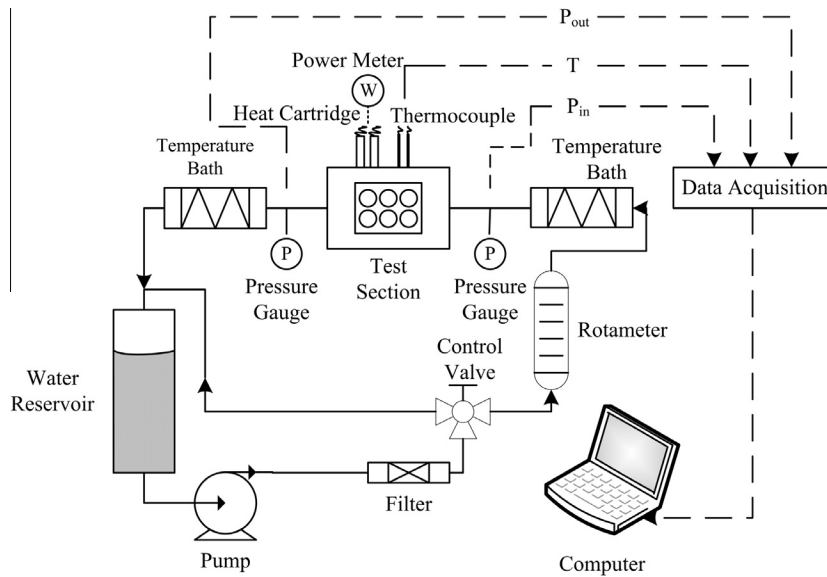


Fig. 8. Schematic diagram of the experimental setup.

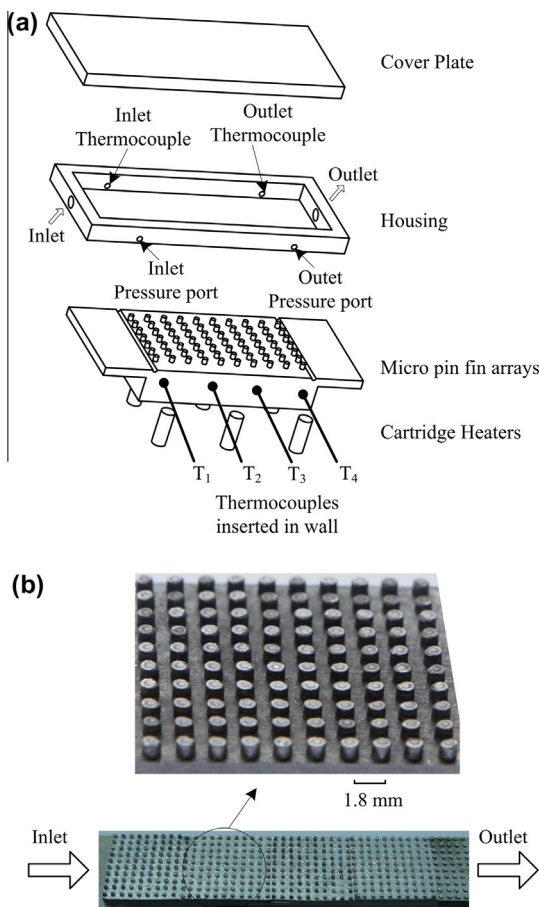


Fig. 9. The structure of the MPFAR: (a) test section construction; (b) micro-pin-fin arrays.

provides a low-cost and high-efficiency approach for the fabrication of micro-pin-fin arrays [4]. The micro-pin-fin arrays had a top area with a width (W) of 1.8 cm and a length (L) of 9 cm. An array of 500 micro-pin-fins in line was fabricated on the top surface. The top view with key dimensions of the micro-pin-fin arrays is

shown in Fig. 1(b). Below the top surface, four type-J thermocouples (T_1 – T_4) were uniformly inserted into the cubes at the same height along the flow direction to measure the streamwise temperature distribution in the substrate, as shown in Fig. 9(a). On the bottom surface of the pin-fin arrays, six holes were drilled to accommodate the cartridge heaters, which were controlled by an adjustable power meter. Different constant heat fluxes could be supplied by shifting the adjustable power meter.

In the present study, different tip clearances were obtained through milling different depths of the central portion of the housing where the pin-fin arrays were inserted. The tip clearances were set as 200 μm , 500 μm and 1000 μm , respectively. The housing contained deep plenums both at the inlet and outlet in the micro-pin-fin arrays. Two thermocouples (HUALIN, J-001) and two absolute pressure transducers (LEADER, LDN500) connected to the inlet and outlet deep plenums were used to measure the temperature of the inlet and outlet, and the pressure drop across the micro-pin-fin arrays, respectively. The accuracy of the pressure transducer was estimated to be better than 1%. The uncertainty in all thermocouple measurements was less than ± 0.1 K and its maximum error was 2%.

The cover plate and the housing with different depth holes were bolted on the top of the micro-pin-fin arrays to form a closed flow passage with different tip clearances. To avoid fluid leakage, a shallow groove was machined on the top surface of the housing and a flexible graphite gasket was used to fill the groove. After the test section was assembled, all the surfaces were wrapped by multiple layers of asbestos fiber to reduce heat loss.

The data collection and processing unit consisted of thermocouples, a flow rotameter and pressure transducers. The readings of the thermocouples and pressure transducers were recorded and processed by an NI data acquisition/control system based on LABVIEW software.

4.2. Data reduction

To conduct a heat transfer and pressure drop test, the inlet temperature T_{in} and velocity u_{in} of the fluid entering the test section were set to match those used in the simulation.

The amount of heat transferred to the micro-pin-fin arrays is equal to the measured change in enthalpy of the water flow,

$$Q = c_p \dot{m} (T_{\text{out}} - T_{\text{in}}) \quad (24)$$

After the flow reached steady state conditions at the desired inlet temperature and flow rate, readings from the thermocouples and pressure transducers were recorded at 0.5 s intervals for 4 min. Then the temperature and pressure data used in this study was obtained by time-averaging the recorded readings over the 4 min measurement period. Every experiment was carried out and measured in the same way for four times to insure the repeatability of the results. The max errors of temperature and pressure drop measurements are 3.5% and 2.7%, respectively. Readings from the flow rotameter and power meter were recorded manually.

Heat loss from the test section is evaluated from the difference between the total electrical power delivered by the cartridge heaters and the measured enthalpy change of the water flow,

$$Q_{\text{loss}} = P_W - c_p \dot{m} (T_{\text{out}} - T_{\text{in}}) \quad (25)$$

As multiple layers of asbestos fiber were wrapped to reduce heat loss, it is found that the heat loss just ranged from 2.1% to 5.8%, which is not significant.

The effective heat flux q applied to the base of the micro-pin-fin arrays is defined as:

$$q = \frac{Q}{A_t} \quad (26)$$

where A_t is the total area of the base of the micro-pin-fin array.

The average heat transfer coefficient h_{ave} is evaluated from

$$qA_t = h_{\text{ave}}(T_{w,\text{ave}} - T_{f,\text{ave}})[(A_t - NA_c) + \eta_{\text{fin}}NA_{\text{fin}}] \quad (27)$$

Eq. (27) is derived from an energy balance between the heat input to the micro-pin-fin arrays and the heat removed from them by the water. $T_{w,\text{ave}}$ represents the average temperature of the channel bottom, and $T_{f,\text{ave}}$ represents the average temperature of the water.

$T_{w,\text{ave}}$ is calculated from

$$T_{w,\text{ave}} = \frac{T_1 + T_2 + T_3 + T_4}{4} - \frac{ql_d}{k_s} \quad (28)$$

$T_{f,\text{ave}}$ is defined as the average of the measured inlet and outlet temperatures of the water,

$$T_{f,\text{ave}} = \frac{T_{\text{in}} + T_{\text{out}}}{2} \quad (29)$$

η_{fin} represents the fin efficiency of the performance as a heat transfer enhancement structure,

$$\eta_{\text{fin}} = \frac{\tanh(m_{\text{fin}}h_f)}{m_{\text{fin}}h_f} \quad (30)$$

and m_{fin} is the fin efficiency coefficient

$$m_{\text{fin}} = \sqrt{\frac{h_{\text{ave}}P_{\text{fin}}}{k_sA_c}} \quad (31)$$

The value of the h_{ave} can be calculated iteratively from Eq. (27) and then the average Nusselt number Nu_{ave} is determined from Eq. (14). The pressure drop ΔP was measured by the inlet and outlet pressure transducers and is calculated using Eq. (15). Then the friction factor f is determined from Eq. (16).

4.3. Comparing results from experimentation and simulation

In this study, the heat transfer and pressure drop experiments were carried out in the MPFAR and the tip clearances were $h_c/h_f = 1.2, 1.5$ and 2. Comparisons of the simulated average heat transfer coefficient and average Nusselt number with the experimental results are shown in Fig. 10(a) and (b), respectively. Fig. 10(a) plots the variation in the average heat transfer

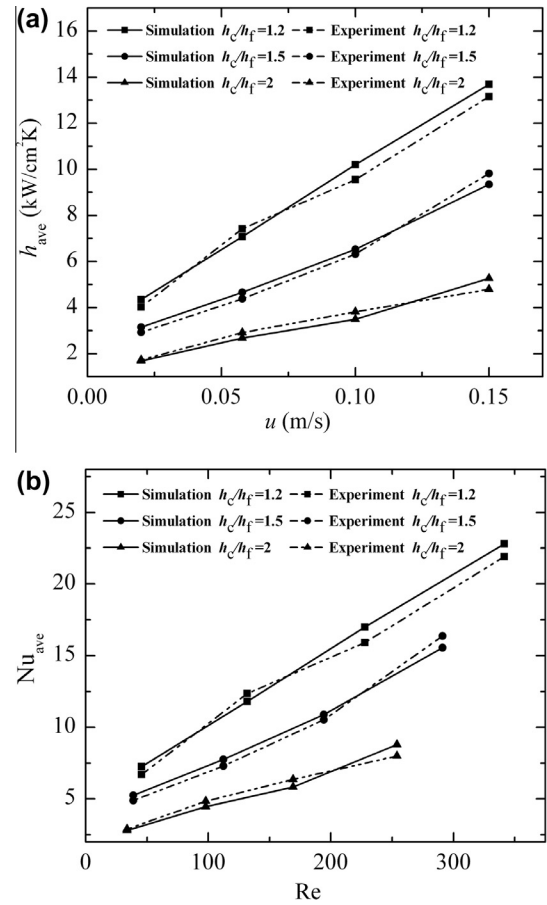


Fig. 10. Comparisons between simulated and experimental results: (a) average heat transfer coefficient vs. inlet velocity; (b) average Nusselt number vs. Reynolds number.

the inlet velocity for the different tip clearances of $h_c/h_f = 1.2, 1.5$ and 2. Fig. 10(b) shows the relationship between the average Nusselt number and the Reynolds number. The values of the average heat transfer coefficient and average Nusselt number obtained from the numerical simulation by CFD agree well with those from experimentation, with a maximum error of 9.07%.

The simulated pressure drop and friction factor are compared with the experimental results in Fig. 11(a) and (b). Fig. 11(a) describes the relationship between the pressure drop and the inlet velocity under different tip clearances of $h_c/h_f = 1.2, 1.5$ and 2. The maximum error of the pressure drop between the simulated and the experimental results is less than 10.59% when the inlet velocity is 0.02 m/s. The variation in the friction factor with the Reynolds number is plotted in Fig. 11(b). The simulated friction factors are reasonably consistent with the experimental results.

Meanwhile, results from the numerical analysis and experimental tests show an initial increasing of heat transfer capability at low level of tip clearance (between 0% and 50%). One reason is that the pin-fins become exposed to the cooling fluid and provide additional heat transfer area to the pin-fin arrays, another one is that the top surface of the pin-fins can contribute to the heat transfer enhancement by regeneration of the thermal boundary layer. However, both of the heat transfer on the top surface of the pin-fins and the heated base bottom decrease at high level of tip clearance (above 50%), therefore, the heat transfer enhancement decreases with the increase of the tip clearance at high level of tip clearance. In addition, introducing tip clearance results in a decrease in form drag and a relatively smaller increase in shear stress, so the friction factor decreases with the increase of the tip clearance in this study.

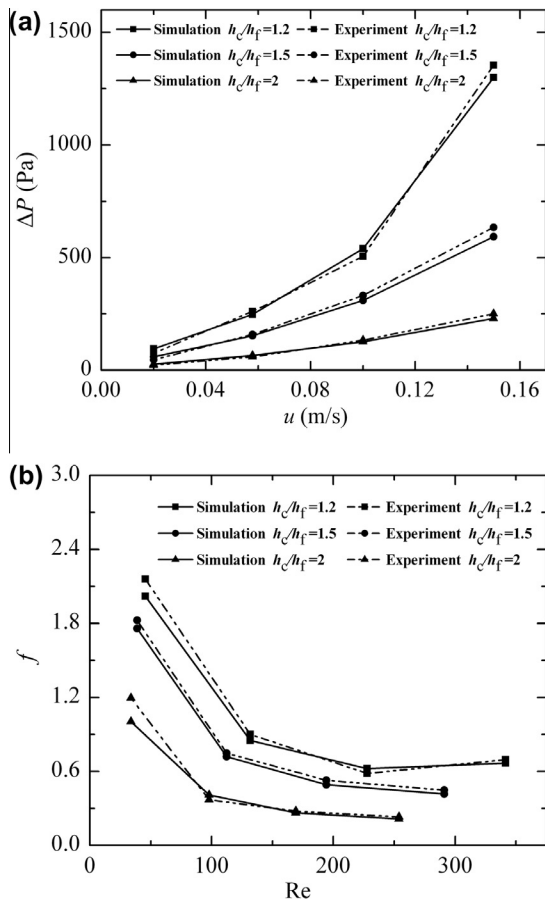


Fig. 11. Comparisons between simulated and experimental results: (a) pressure drop vs. inlet velocity; (b) friction factor vs. Reynolds number.

5. Conclusions

In this study, the MPFAR with different tip clearances was studied to investigate the effect of tip clearance on the heat transfer and pressure drop performance at low Reynolds numbers. A computational model of the MPFAR with or without tip clearance was developed, and the thermal and hydrodynamic performance was studied by numerical analysis. The heat transfer performance decreased with increasing tip clearance, while the pressure drop performance was improved. The heat transfer performance in the MPFAR with a tip clearance of $h_c/h_f = 1.1$ or 1.2 was much better than that without tip clearance when the Reynolds number ranged from 33 to 350. Previous heat transfer and friction factor correlations were examined and found to overpredict the present average Nusselt number and friction factor data in the micro reactor with micro-pin-fins arrays. Therefore, new heat transfer and friction factor correlations accounting for the effect of tip clearance at low Reynolds number were developed. The terms $(h_c/h_f)^{-1.41}$ and $(h_c/h_f)^{-1.64}$, accounting for the effect of tip clearance, were introduced into the new heat transfer and friction factor correlations, respectively. The new correlations indicated a strong dependence of the average Nusselt number and friction factor on the tip clearance. Finally, a series of experiments was conducted to validate the rationality of the numerical analysis. The simulated results agreed well with the experimental results. The present study expands the understanding of the heat transfer and pressure drop performance, and provides a valuable approach for designing and optimizing the tip clearance of the MPFAR.

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